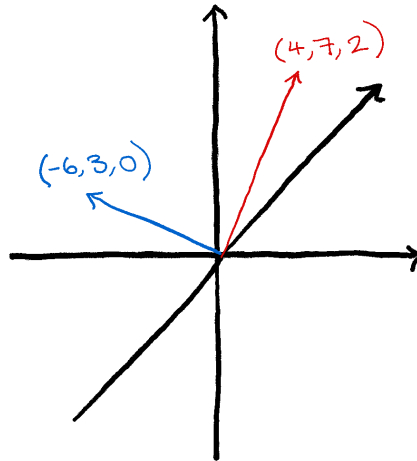


Week 1

1.1 Euclidean Space

In this course, we are concerned with the analysis of functions between finite dimensional vector spaces over the real numbers $\mathbb{R}$. For some fixed dimension $n \in \mathbb{N}$, up to (linear) isomorphism such a vector space is n -dimensional Euclidean space $\mathbb{R}^n$.

$$\mathbb{R}^n = \underbrace{\mathbb{R} \times \cdots \times \mathbb{R}}_{n\text{-times}} = \{(x_1, \dots, x_n) : x_i \in \mathbb{R}, \quad \forall i \in \{1, \dots, n\}\}.$$



Notation: In these printed notes we will use an overhead arrow $\vec{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ or just standard lettering $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ to denote a vector, with subscripts denoting the component of such a vector in Cartesian coordinates. In some cases, we will also use capital letters $A, B, C, \dots$ to represent points within $\mathbb{R}^n$, and use $\overrightarrow{AB}$ to denote the vector starting from point A and ending at point B .

1.1.1 Basic Operations on Vectors

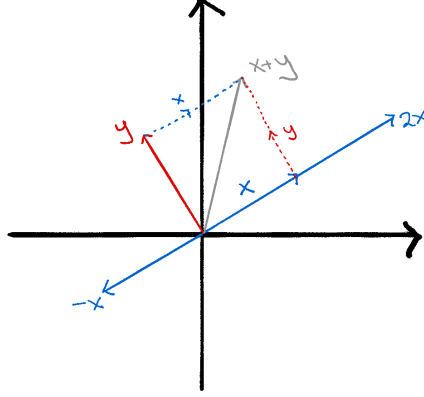
Let $\vec{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$, $\vec{y} = (y_1, \dots, y_n) \in \mathbb{R}^n$ be a pair of vectors and $\lambda \in \mathbb{R}$ a scalar.

- Equality: $\vec{x} = \vec{y} \iff x_i = y_i$, for all $i \in \{1, \dots, n\}$.

‘Two vectors are the same if and only if all of their components agree.’

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- Addition: $\vec{x} + \vec{y} = (x_1 + y_1, \dots, x_n + y_n)$.
‘Vectors are summed component wise.’
- Scalar multiplication: $\lambda \vec{x} = (\lambda x_1, \dots, \lambda x_n)$.
‘Scaling a vector scales all of its components by the same amount.’



1.1.2 Length and the Dot Product

As well as the usual vector space structure of $\mathbb{R}^n$, we also equip it with its natural inner product, known as the *dot product*.

Definition 1.1. For any pair of vectors $\vec{x}, \vec{y} \in \mathbb{R}^n$ we define their dot product to be

$$\vec{x} \cdot \vec{y} := \sum_{i=1}^n x_i y_i = x_1 y_1 + \dots + x_n y_n \in \mathbb{R}.$$

For any vector $\vec{x} \in \mathbb{R}^n$ we define its (Euclidean) length $\|\vec{x}\|$ via the equation

$$\|\vec{x}\|^2 := \vec{x} \cdot \vec{x} = x_1^2 + \dots + x_n^2.$$

We note that the dot product is a map $\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ which takes a pair of vectors and returns a scalar value.

Remark. In dimension $n \leq 3$, $\|\vec{x}\|$ agrees with the usual notion of length you calculate by applying the Pythagorean theorem.

Lemma 1. Let $\vec{x}, \vec{y}, \vec{z} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$. The dot product enjoys the following properties.

- a) $\vec{x} \cdot \vec{y} = \vec{y} \cdot \vec{x}$.
- b) $(\vec{x} + \vec{y}) \cdot \vec{z} = \vec{x} \cdot \vec{z} + \vec{y} \cdot \vec{z}$.

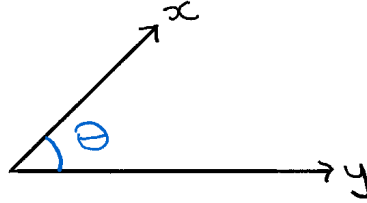
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c) $(\lambda \vec{x}) \cdot \vec{y} = \lambda(\vec{x} \cdot \vec{y}) = \vec{x} \cdot (\lambda \vec{y})$.

d) $\vec{x} \cdot \vec{x} \geq 0$, with equality iff $\vec{x} = 0$.

In the case where $n = 2$ or $n = 3$, we furthermore have:

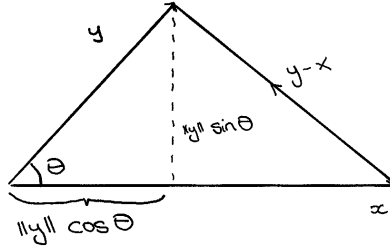
e) If $\vec{x} \neq 0$ and $\vec{y} \neq 0$, then $\vec{x} \cdot \vec{y} = \|\vec{x}\| \|\vec{y}\| \cos \theta$, where θ is the size of the angle formed between the vectors $\vec{x}$ and $\vec{y}$.



Proof. Properties a)-d) follow trivially from the definitions and are left as an exercise. To show e) we first consider the case where $\theta \in [0, \frac{\pi}{2}]$. By the definition of the dot product, we have

$$\begin{aligned} \|\vec{y} - \vec{x}\|^2 &= (\vec{y} - \vec{x}) \cdot (\vec{y} - \vec{x}) \\ &= \vec{y} \cdot \vec{y} - \vec{x} \cdot \vec{y} - \vec{y} \cdot \vec{x} + \vec{x} \cdot \vec{x} \\ &= \|\vec{y}\|^2 + \|\vec{x}\|^2 - 2\vec{x} \cdot \vec{y}. \end{aligned}$$

Furthermore, by the Pythagorean theorem



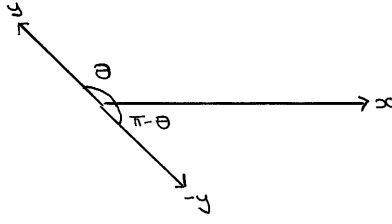
$$\begin{aligned} \|\vec{y} - \vec{x}\|^2 &= (\|\vec{x}\| - \|\vec{y}\| \cos \theta)^2 + (\|\vec{y}\| \sin \theta)^2 \\ &= \|\vec{y}\|^2 + \|\vec{x}\|^2 - 2\|\vec{x}\| \|\vec{y}\| \cos \theta. \end{aligned}$$

Comparing the two gives the result for $\theta \in [0, \frac{\pi}{2}]$. Finally, if $\theta \in (\frac{\pi}{2}, \pi]$, we consider instead the vectors $\vec{x}$ and $-\vec{y}$, which now form an angle of size $\pi - \theta \in [0, \frac{\pi}{2}]$. We may then apply the above result to conclude that

$$\vec{x} \cdot (-\vec{y}) = \|\vec{x}\| \|\vec{y}\| \cos(\pi - \theta).$$

Since $\vec{x} \cdot (-\vec{y}) = -(\vec{x} \cdot \vec{y})$, $\|\vec{y}\| = \|\vec{y}\|$, and $\cos(\pi - \theta) = -\cos(\theta)$, the result follows. $\square$

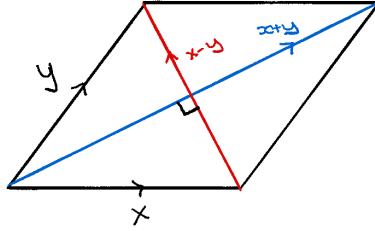
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Remark. For any two vectors $\vec{x}, \vec{y} \in \mathbb{R}^n$ which form an angle of size θ ,

$$\vec{x} \text{ and } \vec{y} \text{ are orthogonal} \iff \theta = \frac{\pi}{2} \iff \cos(\theta) = 0 \iff \vec{x} \cdot \vec{y} = 0.$$

The following is a simple lemma which states that the diagonals of a parallelogram are orthogonal to each other if and only if the parallelogram is in fact a rhombus.



Lemma 2. For any two vectors $\vec{x}, \vec{y} \in \mathbb{R}^n$, $\vec{x} + \vec{y}$ is orthogonal to $\vec{x} - \vec{y}$ if and only if $\vec{x}$ and $\vec{y}$ have the same length.

Proof. From the previous remark, $\vec{x} + \vec{y}$ and $\vec{x} - \vec{y}$ are orthogonal iff $(\vec{x} + \vec{y}) \cdot (\vec{x} - \vec{y}) = 0$, but a direct calculation leads to

$$(\vec{x} + \vec{y}) \cdot (\vec{x} - \vec{y}) = \vec{x} \cdot \vec{x} + \vec{x} \cdot \vec{y} - \vec{x} \cdot \vec{y} - \vec{y} \cdot \vec{y} = \|\vec{x}\|^2 - \|\vec{y}\|^2,$$

from which the result follows immediately. □

As an application of the previous lemma, we give a simple proof that for any triangle lying on a circle with one edge given by a diameter, the opposite angle is $\frac{\pi}{2}$.

If A, B, C denote the vertices of the triangle with A and B lying on the diameter and O denoting the centre of the circle, then $\overrightarrow{AO} = -\overrightarrow{BO}$, and hence

$$\overrightarrow{AC} = \overrightarrow{OC} + \overrightarrow{AO}, \quad \overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{AO}.$$

Since the lengths of $\overrightarrow{OC}$ and $\overrightarrow{AO}$ are the same (the radius of the circle), we can apply the previous lemma to conclude that $\overrightarrow{AC}$ and $\overrightarrow{BC}$ are orthogonal.

1.1.3 Cauchy-Schwarz and the Triangle Inequality.

The angle θ between two vectors $\vec{x}$ and $\vec{y}$ is defined for $\vec{x}, \vec{y} \in \mathbb{R}^2$ or $\mathbb{R}^3$ using elementary geometry. We would like to define the angle for vectors in higher dimensional space. One way to do this would be to use part e) of Lemma 1, and *define* the angle as

$$\theta = \cos^{-1} \left(\frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|} \right).$$

However, for this to make sense, we need to ensure that the size of $\frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|}$ is less than 1. This is precisely what the following important inequality says.

Lemma 3 (Cauchy-Schwarz inequality). *For any $\vec{x}, \vec{y} \in \mathbb{R}^n$,*

$$|\vec{x} \cdot \vec{y}| \leq \|\vec{x}\| \|\vec{y}\|. \quad (1.1)$$

Moreover, equality holds if and only if $\vec{x}$ and $\vec{y}$ are parallel. That is, $\vec{x} = \lambda \vec{y}$ or $\vec{y} = \lambda \vec{x}$ for some $\lambda \in \mathbb{R}$.

Remark. *For $n \leq 3$, this follows immediately from the equality $\vec{x} \cdot \vec{y} = \|\vec{x}\| \|\vec{y}\| \cos \theta$.*

Proof. If $\vec{y} = 0 \in \mathbb{R}^n$, then the result is trivially true with $\vec{y} = 0 \cdot \vec{x}$. Therefore, we may assume throughout the proof that $\vec{y} \neq 0$. For any $\lambda \in \mathbb{R}$, we consider the quantity

$$0 \leq \|\vec{x} - \lambda \vec{y}\|^2 := (\vec{x} - \lambda \vec{y}) \cdot (\vec{x} - \lambda \vec{y}) = \|\vec{x}\|^2 - 2\lambda \vec{x} \cdot \vec{y} + \lambda^2 \|\vec{y}\|^2. \quad (1.2)$$

As $\vec{y} \neq 0$, we may choose $\lambda := \frac{\vec{x} \cdot \vec{y}}{\|\vec{y}\|^2} \in \mathbb{R}$ in (1.2), which simplifies to

$$\|\vec{x}\|^2 - \frac{|\vec{x} \cdot \vec{y}|^2}{\|\vec{y}\|^2} \geq 0.$$

Rearranging, we can conclude (1.1) holds. Moreover, equality in (1.1) holds iff $\|\vec{x} - \lambda \vec{y}\| = 0$ for our choice of λ , which happens iff $\vec{x}$ and $\vec{y}$ are parallel. $\square$

Remark. *Let $\vec{x}, \vec{y} \in \mathbb{R}^n$ be non-zero vectors. The Cauchy-Schwarz inequality now implies that*

$$-1 \leq \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|} \leq 1.$$

Thus, we can now define

$$\theta = \cos^{-1} \left(\frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|} \right) \in [0, \pi],$$

to be the angle between $\vec{x}$ and $\vec{y}$ in all dimensions.

Example 4. For the pair of vectors $\vec{x} = (1, 5, 2, 3, 4), \vec{y} = (3, 3, 1, 1, 0) \in \mathbb{R}^5$, we have

$$\vec{x} \cdot \vec{x} = 55, \quad \vec{y} \cdot \vec{y} = 20, \quad \vec{x} \cdot \vec{y} = 23.$$

Therefore, the angle between them is $\theta = \arccos \left(\frac{23}{\sqrt{11}} \right)$.

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Just like for the absolute value in $\mathbb{R}$, we have the following triangle inequality holds for the length of vectors in $\mathbb{R}^n$.

Lemma 5 (Triangle Inequality). *For any $\vec{x}, \vec{y} \in \mathbb{R}^n$,*

$$\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|. \quad (1.3)$$

Moreover, equality holds if and only if $\vec{x} = \lambda \vec{y}$ or $\vec{y} = \lambda \vec{x}$ for some $\lambda \geq 0$.

Proof. By expanding the dot product and using the Cauchy-Schwarz inequality

$$\begin{aligned} \|\vec{x} + \vec{y}\|^2 &= \|\vec{x}\|^2 + 2\vec{x} \cdot \vec{y} + \|\vec{y}\|^2 \\ &\leq \|\vec{x}\|^2 + 2|\vec{x} \cdot \vec{y}| + \|\vec{y}\|^2 \\ &\leq \|\vec{x}\|^2 + 2\|\vec{x}\|\|\vec{y}\| + \|\vec{y}\|^2 \\ &= (\|\vec{x}\| + \|\vec{y}\|)^2, \end{aligned}$$

which is precisely (1.3). Moreover, we have equality in (1.3) iff we have equality in Cauchy-Schwarz and $\vec{x} \cdot \vec{y} = |\vec{x} \cdot \vec{y}|$, which happens iff $\vec{x}$ is parallel to $\vec{y}$ with $\lambda \geq 0$. $\square$

1.1.4 The Cross Product

For this section, we restrict our attention to the case $n = 3$.

Before stating the definition of the cross product of two vectors in $\mathbb{R}^3$, we recall that the determinant of a 2×2 matrix is

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc,$$

and for a 3×3 matrix

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}.$$

Definition 1.2. *For any two vectors $\vec{x} = (x_1, x_2, x_3), \vec{y} = (y_1, y_2, y_3) \in \mathbb{R}^3$, we define the cross product to be*

$$\vec{x} \times \vec{y} := \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} = \begin{vmatrix} x_2 & x_3 \\ y_2 & y_3 \end{vmatrix} \hat{i} - \begin{vmatrix} x_1 & x_3 \\ y_1 & y_3 \end{vmatrix} \hat{j} + \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} \hat{k} \in \mathbb{R}^3, \quad (1.4)$$

where $\hat{i} = (1, 0, 0)$, $\hat{j} = (0, 1, 0)$ and $\hat{k} = (0, 0, 1)$.

Example 6. If $\vec{x} = (2, 3, 4)$ and $\vec{y} = (1, 2, 3)$, then

$$\begin{aligned} \vec{x} \times \vec{y} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 1 & 2 & 3 \end{vmatrix} = \begin{vmatrix} 3 & 4 \\ 2 & 3 \end{vmatrix} \hat{i} - \begin{vmatrix} 2 & 4 \\ 1 & 3 \end{vmatrix} \hat{j} + \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} \hat{k} \\ &= \hat{i} - 2\hat{j} + \hat{k} \\ &= (1, -2, 1). \end{aligned}$$

Remark.

$$\begin{aligned} \hat{i} \times \hat{i} &= 0, & \hat{i} \times \hat{j} &= \hat{k}, & \hat{i} \times \hat{k} &= -\hat{j}, \\ \hat{j} \times \hat{i} &= -\hat{k}, & \hat{j} \times \hat{j} &= 0, & \hat{j} \times \hat{k} &= \hat{i}, \\ \hat{k} \times \hat{i} &= \hat{j}, & \hat{k} \times \hat{j} &= -\hat{i}, & \hat{k} \times \hat{k} &= 0. \end{aligned}$$

The cross product satisfies the **right hand rule**.

1.1.5 Properties of the Cross Product

Let $\vec{x}, \vec{y}, \vec{z} \in \mathbb{R}^3$ and $\alpha, \beta \in \mathbb{R}$.

- a) $\vec{x} \times \vec{y} = -\vec{y} \times \vec{x}$
- b) $(\alpha\vec{x} + \beta\vec{y}) \times \vec{z} = \alpha(\vec{x} \times \vec{z}) + \beta(\vec{y} \times \vec{z})$.
- c) $(\vec{x} \times \vec{y}) \cdot \vec{x} = (\vec{x} \times \vec{y}) \cdot \vec{y} = 0$, so $\vec{x} \times \vec{y}$ is orthogonal to the plane spanned by $\vec{x}$ and $\vec{y}$.
- d) If θ denotes the angle between $\vec{x}$ and $\vec{y}$, then

$$\|\vec{x} \times \vec{y}\| = \|\vec{x}\| \|\vec{y}\| \sin \theta,$$

which is equal to the area of the parallelogram generated by $\vec{x}$ and $\vec{y}$.

Proof. Properties a)-c) follow directly from the definition and are left as an Exercise. To show d) we begin by expanding the term $(\vec{x} \times \vec{y}) \cdot (\vec{x} \times \vec{y})$, and find that

$$\begin{aligned} \|\vec{x} \times \vec{y}\|^2 &= \|\vec{x}\|^2 \|\vec{y}\|^2 - (\vec{x} \cdot \vec{y})^2 \\ &= \|\vec{x}\|^2 \|\vec{y}\|^2 (1 - \cos^2(\theta)) \\ &= \|\vec{x}\|^2 \|\vec{y}\|^2 \sin^2(\theta). \end{aligned}$$

Note that the area is given by the base $\|\vec{x}\|$ times the height $\|\vec{y}\| \sin \theta$. □

Remark. In the case that $\vec{x}, \vec{y} \in \mathbb{R}^2$, we see that

$$\vec{x} \times \vec{y} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x_1 & x_2 & 0 \\ y_1 & y_2 & 0 \end{vmatrix} = \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} \hat{k}, \quad (1.5)$$

and so $\|\vec{x} \times \vec{y}\| = |x_1 y_2 - x_2 y_1|$.

1.1.6 Triple Product

For any $\vec{x}, \vec{y}, \vec{z} \in \mathbb{R}^3$, we define their *triple product* as $(\vec{x} \times \vec{y}) \cdot \vec{z} \in \mathbb{R}$.

Unravelling the definition, we see that

$$(\vec{x} \times \vec{y}) \cdot \vec{z} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} \cdot \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{vmatrix} z_1 & z_2 & z_3 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} = \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix}. \quad (1.6)$$

It follows from (1.6) that

$$(\vec{x} \times \vec{y}) \cdot \vec{z} = (\vec{y} \times \vec{z}) \cdot \vec{x} = (\vec{z} \times \vec{x}) \cdot \vec{y} = -(\vec{y} \times \vec{x}) \cdot \vec{z} = -(\vec{z} \times \vec{y}) \cdot \vec{x} = -(\vec{x} \times \vec{z}) \cdot \vec{y}$$

The triple product has the following geometric interpretation.

Lemma 7. $|(\vec{x} \times \vec{y}) \cdot \vec{z}|$ is the volume of the parallelepiped spanned by $\vec{x}$, $\vec{y}$ and $\vec{z}$.

Proof. Let α denote the angle formed between the vector $\vec{x} \times \vec{y}$ and $\vec{z}$.

Note that, after possibly replacing $\vec{x} \times \vec{y}$ with $\vec{y} \times \vec{x}$, we may assume that $\alpha \in [0, \frac{\pi}{2}]$.

Since $(\vec{x} \times \vec{y}) \cdot \vec{z} = \|\vec{x} \times \vec{y}\| \|\vec{z}\| \cos \alpha$, and our parallelepiped has base area $\|\vec{x} \times \vec{y}\|$ and height $\|\vec{z}\| \cos \alpha$, the result follows. $\square$

Example 8. Consider the parallelepiped spanned by the vectors $(1, 1, 0)$, $(0, 1, 0)$, $(1, 1, 1) \in \mathbb{R}^3$. The triple product of these three vectors is

$$\begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{vmatrix} = 1,$$

and therefore the parallelepiped has unit volume.

1.1.7 Linear Dependence

In this subsection, we define what it means for a collection of vectors to be linearly independent.

Definition 1.3. Let $\vec{v}_1, \dots, \vec{v}_k \in \mathbb{R}^n$ be a collection of vectors. We say that these vectors are linearly dependent if there exist scalars $\alpha_1, \dots, \alpha_k \in \mathbb{R}^k$ such that

$$\alpha_1 \vec{v}_1 + \alpha_2 \vec{v}_2 + \dots + \alpha_k \vec{v}_k = 0.$$

Otherwise, we say that the vectors are linearly independent.

Another way to state this definition is that for linearly independent $v_1, \dots, v_k \in \mathbb{R}^n$, the only scalars $\alpha_i \in \mathbb{R}$ such that

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k = 0,$$

are $\alpha_i = 0$, for all $i \in \{1, \dots, k\}$.

Example 9. For two vectors $v_1, v_2 \in \mathbb{R}^n$, v_1 and v_2 are linearly dependent if either $v_1 = rv_2$ or $rv_1 = v_2$ for some $r \in \mathbb{R}$, i.e. v_1 and v_2 are parallel.

Example 10. Consider the three vectors $(0, 2, 1)$, $(1, -4, -1)$, $(1, 0, 1) \in \mathbb{R}^3$. We note that

$$2 \cdot (0, 2, 1) + (1, -4, -1) - (1, 0, 1) = 0,$$

and so these three vectors are linearly dependent. In particular, these three vectors all lie in the same plane.

Remark. The triple product $(\vec{x} \times \vec{y}) \cdot \vec{z} = 0$ iff the volume of the parallelepiped vanishes iff $\{\vec{x}, \vec{y}, \vec{z}\}$ are linearly dependent.

1.2 Linear Objects in $\mathbb{R}^n$

1.2.1 Lines in $\mathbb{R}^2$

A line L in $\mathbb{R}^2$ is determined by one of the following:

- (i) Two points A, B on the line
- (ii) A point A on the line and a tangent vector $\vec{v}$ to the line
- (iii) A point A on the line and a normal vector $\vec{n}$ to the line

Recall that if $\vec{v} = (x, y)$ is a tangent vector to the line, then $\vec{n} = (-y, x)$ is a normal vector to the line, so we may convert between (ii) and (iii) easily. If instead we have description (i), i.e. two points $A, B \in L$, then a tangent vector is given by $\vec{v} = \overrightarrow{AB} = B - A$.

Let $L \subseteq \mathbb{R}^2$ be a line. Suppose $A \in L$ is a point on the line and $v \in \mathbb{R}^n$ is the direction of L ; that is, $v \in L - A$. Then a *parametric form* of the line is given by

$$\vec{x} = A + t\vec{v},$$

where the real number $t \in \mathbb{R}$ is called a *parameter*. As a set, the line is $L = \{A + t\vec{v} : t \in \mathbb{R}\}$.

The *equation* determining the line L is given by:

$$\vec{x} \cdot \vec{n} = A \cdot \vec{n}.$$

This is the usual equation in terms of x and y that we are familiar with from elementary mathematics.

1.2.2 Lines in $\mathbb{R}^3$

In $\mathbb{R}^3$, (i) and (ii) still determine a unique line. However, (iii) is no longer valid, as there are *many* lines through a point A which are normal to a vector $\vec{n}$! For example, in $\mathbb{R}^3$, if $A = (0, 0, 0)$ and $\vec{n} = (0, 0, 1)$, then note that the line through A with tangent vector $\vec{v} = (\cos(\theta), \sin(\theta), 0)$ is perpendicular to $\vec{n}$, for *any* $\theta \in [0, 2\pi)$.

How can we express L in terms of equations? We start with a parametric form $\vec{x} = A + t\vec{v}$, and write it in coordinates:

$$x = a_1 + tv_1, \quad y = a_2 + tv_2, \quad z = a_3 + tv_3.$$

Then we can rearrange each in terms of t and equate them to find that:

$$\frac{x - a_1}{v_1} = \frac{y - a_2}{v_2} = \frac{z - a_3}{v_3}.$$

So in fact, a line in $\mathbb{R}^3$ is determined by *two* equations, not one.

Example 11. Suppose $L \subseteq \mathbb{R}^3$ is the line passing through the points $A = (1, 2, 3)$ and $B = (-1, 3, 5)$.

In this case, we may take

$$\vec{v} = \overrightarrow{AB} = (-1, 3, 5) - (1, 2, 3) = (-2, 1, 2),$$

so that a parametric form of the line is

$$\vec{x} = (1, 2, 3) + t \cdot (-2, 1, 2).$$

To find the equation for the line L , we solve for t :

$$\frac{1-x}{2} = \frac{y-2}{1} = \frac{z-3}{2} (=t).$$

Remark. The parametric form of a line is not unique.

- Choosing a different initial point on the line corresponds to translating the value of t by a fixed amount:

In the previous example, we have chosen $A = (-1, 3, 5)$, giving the parametric form of the line

$$\vec{x} = (-1, 3, 5) + s \cdot (-2, 1, 2).$$

This is the same as the original parameterisation by setting $t = s - 1$.

- Choosing a stretched version of our direction vector corresponds to scaling the value of t by a fixed amount:

In the previous example, we have chosen $v = (4, -2, -4)$, giving the parametric form of the line

$$\vec{X} = (1, 2, 3) + s \cdot (4, -2, -4).$$

This may be identified with the original parameterisation by setting $t = -2s$.

1.2.3 Lines in $\mathbb{R}^n$

A line L in $\mathbb{R}^n$ is still determined by (i) and (ii), and we can write it in parametric form $\vec{x} = A + t\vec{v}$ as before. We can use the same argument as above to express the line in terms of $(n-1)$ equations:

$$\frac{x_1 - a_1}{v_1} = \dots = \frac{x_n - a_n}{v_n}.$$

1.2.4 Planes in $\mathbb{R}^3$

A plane $P \subseteq \mathbb{R}^3$ is determined by

- (i) Three non-colinear points on P ;
- (ii) A point on P , and two linearly independent tangent vectors $\vec{u}, \vec{v}$ in P ;
- (iii) A point on P and a normal vector $\vec{n}$ to P .

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For case ii), if $A \in P$ and $\vec{u}, \vec{v} \in \mathbb{R}^3$ are linearly independent tangent vectors in P , then we have the following parametric form of the plane:

$$\vec{x} = A + t\vec{u} + s\vec{v}.$$

For case iii), if $A = (a_1, a_2, a_3) \in P$ and $\vec{n} = (n_1, n_2, n_3)$ is a normal vector to P , then for any $\vec{x} \in \mathbb{R}^3$ we see that

$$\vec{x} \in P \iff \vec{x} - A \perp \vec{n} \iff (\vec{x} - A) \cdot \vec{n} = 0 \iff \vec{x} \cdot \vec{n} = A \cdot \vec{n},$$

and hence the equation of the plane is

$$n_1x_1 + n_2x_2 + n_3x_3 = \underbrace{a_1n_1 + a_2n_2 + a_3n_3}_{\in \mathbb{R}},$$

where the right hand side is just a constant.

In particular, for any non-zero vector $(a, b, c) \in \mathbb{R}^3$, the set of points $(x, y, z) \in \mathbb{R}^3$ solving the equation

$$ax + by + cz = d, \quad (d \in \mathbb{R}),$$

is a plane with normal vector (a, b, c) . We note that if $(a, b, c) = 0 \in \mathbb{R}^3$, then this equation describes either all of $\mathbb{R}^3$ when $d = 0$, or the empty set when $d \neq 0$.

Example 12. Let P be the plane passing through the points $A = (0, 0, 1)$, $B = (0, 2, 0)$, and $C = (-1, 1, 0)$. To find a parametric form of P , we note that the vectors

$$\vec{AB} = (0, 2, -1), \quad \vec{AC} = (-1, 1, -1),$$

are directions in P , and therefore

$$P = \{(0, 0, 1) + s(0, 2, -1) + t(-1, 1, -1) : s, t \in \mathbb{R}\}.$$

Next we note that the vector

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 2 & -1 \\ -1 & 1 & -1 \end{vmatrix} = (-1, 1, 2),$$

is a normal vector to P , and so the equation of the plane is given by

$$-x + y + 2z = (-1, 1, 2) \cdot (x, y, z) = (-1, 1, 2) \cdot (0, 0, 1) = 2.$$

For two subsets $\mathcal{A}, \mathcal{B} \subseteq \mathbb{R}^n$, we define the distance between these subsets to be the infimum of the distance between pairs of points in these sets

$$\inf_{A \in \mathcal{A}, B \in \mathcal{B}} \|\vec{AB}\|.$$

The following example shows how to calculate the distance between a point and a plane.

Example 13. Let $A = (2, 1, 1)$ and P be the plane given by the equation

$$-x + 2y - z = -4, \quad (1.7)$$

so that P has normal vector $n = (-1, 2, -1)$. Consider the line

$$L := \{A + tn = (2 - t, 1 + 2t, 1 - t) : t \in \mathbb{R}\},$$

and let $B = L \cap P$ be the intersection point of this line and the plane.

Fact: B is the point on P closest to A .

To find B , substitute our formula for a point on L into (1.7)

$$(-1)(2 - t) + 2(1 + 2t) - (1 - t) = -4 \iff t = -\frac{1}{2},$$

and hence $B = (\frac{5}{2}, 0, \frac{3}{2})$. Therefore, the distance between A and P is

$$\|\overrightarrow{AB}\| = \|(1/2, -1, 1/2)\| = \sqrt{6}/2.$$

Exercise: Find the distance between the lines

$$L_1 = \{(-4, 9, -4) + s(4, -3, 0) : s \in \mathbb{R}\},$$

$$L_2 = \{(5, 2, 10) + t(4, 3, 2) : t \in \mathbb{R}\}.$$

Hint: Find $A \in L_1$, $B \in L_2$ such that $\overrightarrow{AB} \perp L_1, L_2$.

Note that lines in $\mathbb{R}^3$ may be viewed as the intersection of two planes:

Example 14. Consider the line

$$x + y + 6z = 6, \quad x - y - 2z = -2.$$

Then, performing Gaussian elimination of this system of equations, we find a general solution is

$$x = 2 - 2t, y = 4 - 4t, z = t, \quad t \in \mathbb{R},$$

which is a parametric form of the line. Conversely, we can eliminate the parameter t from the parametric form to find

$$2x - y = 0, \quad y + 4z = 4,$$

which is another set of equations defining the same line.

Example 15. Given three non-trivial planes in $\mathbb{R}^3$, what is the intersection of those planes?

Case 1: A single unique point. Note that in this case, the normal vectors to the three planes are linearly independent.

Case 2: A line or a plane. Note that in this case, the normal vectors to the three planes are linearly dependent.

Case 3: The empty set. Similar to the previous case, the normal vectors to the three planes are linearly dependent.

1.2.5 General Affine Subspaces

Given a non-zero vector $n \in \mathbb{R}^n$, the solutions $x \in \mathbb{R}^n$ to the equation

$$n \cdot x = \lambda, \quad (\lambda \in \mathbb{R})$$

describes a hyperplane (a dimension $n - 1$ affine subspace) with normal vector n .

To describe a dimension k plane P in $\mathbb{R}^n$, we can either describe it using its

- Parametric form:

$$P = \{q + \sum_{j=1}^k t_j v_j : t_1, \dots, t_k \in \mathbb{R}\},$$

where $q \in P$, $v_1, \dots, v_k$ are linearly independent vectors in the direction of P , and $t_1, \dots, t_k \in \mathbb{R}$ are parameters.

- Equations:

$$\sum_{j=1}^n a_{ij} x_j = c_i, \quad \forall i \in \{1, \dots, n - k\},$$

where $c = (c_1, \dots, c_{n-k}) \in \mathbb{R}^{n-k}$, $A = (a_{ij})$ is a $(n - k) \times n$ matrix with full rank (trivial kernel).

Note that a system of $(n - k)$ equations is equivalent to the intersection of $(n - k)$ hyperplanes. In this language, the condition of the matrix A is equivalent to the normal vectors to the $(n - k)$ hyperplanes being linearly independent.

Exercise: Show that in the parametric form of P , we may choose the direction vectors $v_1, \dots, v_k$ to be orthonormal, i.e.

$$v_i \cdot v_j = \begin{cases} 1 & : i = j \\ 0 & : i \neq j \end{cases}, \quad \forall i, j \in \{1, \dots, k\}.$$

Hint: If you are struggling, look up the Gram-Schmidt algorithm and apply it here.

1.3 Curves

Definition 1.4. Let $I \subseteq \mathbb{R}$ be an interval. A curve in $\mathbb{R}^n$ is a continuous function $x : I \rightarrow \mathbb{R}^n$. That is,

$$\vec{x}(t) = (x_1(t), x_2(t), \dots, x_n(t)), \quad \forall t \in I,$$

where every component function $x_i : I \rightarrow \mathbb{R}$, for $i \in \{1, \dots, n\}$, is continuous.

Example 16. $\vec{x} : [-1, 1) \rightarrow \mathbb{R}^2$, $\vec{x}(t) = (t, t^2)$. This traces a section of the parabola $y = x^2$ starting at the point $(-1, 1)$ and moving in the positive x direction up to (but not including) the point $(1, 1)$.

Example 17. $\vec{x} : \mathbb{R} \rightarrow \mathbb{R}^3$, $x(t) = p + tq$ for some $\vec{p}, \vec{q} \in \mathbb{R}^3$ ($q \neq 0$). This is a parameterisation of a straight line as we saw earlier.

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In the following definition, we restrict our attention to curves whose domain is a closed interval $I = [a, b] \subseteq \mathbb{R}$.

Definition 1.5. A curve $\vec{x} : [a, b] \rightarrow \mathbb{R}^n$ is said to be

- (a) **closed** if $\vec{x}(a) = \vec{x}(b)$.
- (b) **simple** if $\vec{x}(t_1) \neq \vec{x}(t_2)$ for any $a \leq t_1 < t_2 \leq b$, except for possibly $t_1 = a, t_2 = b$.

Theorem 18. Let $\vec{x}(t) = (x_1(t), \dots, x_n(t))$ be a curve in $\mathbb{R}^n$. Then

- $\lim_{t \rightarrow a} \vec{x}(t) = \left(\lim_{t \rightarrow a} x_1(t), \dots, \lim_{t \rightarrow a} x_n(t) \right)$.
- $\vec{x}'(t) = \lim_{h \rightarrow 0} \frac{\vec{x}(t+h) - \vec{x}(t)}{h} = (x'_1(t), \dots, x'_n(t))$, provided the limit exists.

Remark. $\vec{x}'(a)$ is the tangent vector of $\vec{x}$ at $t = a$. If a curve $\vec{x} : I \rightarrow \mathbb{R}^n$ has a well defined tangent vector at every point $t \in I$, then we say the curve is differentiable. If moreover, the map $\vec{x}' : I \rightarrow \mathbb{R}^n$ is continuous (so the derivative is itself a curve), then we say that the original curve is a C^1 -curve.

Example 19. The curve $\vec{x} : \mathbb{R} \rightarrow \mathbb{R}^2$, $\vec{x}(t) = (t, |t|)$ is not differentiable as it has no tangent vector at $t = 0$.

Example 20. The curve $\vec{x} : \mathbb{R} \rightarrow \mathbb{R}^2$, $\vec{x}(t) = (t^2 \sin t^{-1}, 0)$ for $t \neq 0$, and $x(0) = (0, 0)$ is a differentiable curve, but is not a C^1 -curve.

We have the following physical interpretation for derivatives of curves:

Suppose $\vec{x}(t)$ denotes the displacement of a particle at time t . Then $\vec{x}'(t)$ will denote the particles velocity, $\|\vec{x}'(t)\|$ its speed, and $\vec{x}''(t)$ its acceleration at time t .

Example 21. Suppose $\vec{x}(t) = (\cos t, \sin t)$ for $t \in [0, 2\pi]$. Then

$$\begin{aligned}\vec{x}'(t) &= (-\sin t, \cos t) \perp \vec{x}(t), \quad \forall t \in [0, 2\pi], \\ \vec{x}''(t) &= (-\cos t, -\sin t) = -\vec{x}(t), \quad \forall t \in [0, 2\pi],\end{aligned}$$

and $\|\vec{x}'(t)\| \equiv 1$.

Example 22. $\vec{x} : [1, \infty) \rightarrow \mathbb{R}^2$, $\vec{x}(t) = (t^{-1}, t^{-2})$. Then $\lim_{t \rightarrow \infty} \vec{x}(t) = (0, 0)$.

1.3.1 Properties for Derivatives of Curves

Let $\vec{x}, \vec{y} : I \rightarrow \mathbb{R}^n$ be differentiable curves, $\lambda \in \mathbb{R}$, and $f : I \rightarrow \mathbb{R}$ a differentiable function.

- a) $(\vec{x} + \vec{y})'(t) = \vec{x}'(t) + \vec{y}'(t)$;
- b) $(\lambda \vec{x})'(t) = \lambda \cdot \vec{x}'(t)$;
- c) $(f \cdot \vec{x})'(t) = f'(t)\vec{x}(t) + f(t)\vec{x}'(t)$;
- d) $(\vec{x} \cdot \vec{y})'(t) = \vec{x}'(t) \cdot \vec{y}(t) + \vec{x}(t) \cdot \vec{y}'(t)$;
- e) If $n = 3$, $(\vec{x} \times \vec{y})'(t) = \vec{x}'(t) \times \vec{y}(t) + \vec{x}(t) \times \vec{y}'(t)$.

1.3.2 Arclength of a Curve

Suppose $\vec{x} : I \rightarrow \mathbb{R}^n$ is a C^1 -curve.

Definition 1.6. The arclength of the curve $\vec{x}(t)$ for $a \leq t \leq b$ is

$$S := \int_a^b \|\vec{x}'(t)\| dt.$$

Referring back to the physical interpretation of a curve and its derivatives, we are saying that the integral of the speed between two times is equal to the distance travelled.

Alternatively, we could view the arc-length as a limit of the following approximations:

Take $a = t_0 < t_1 < \dots < t_n = b$ for some subdivision of the interval $[a, b]$. Then consider the following sequence of line segments which approximate our curve

$$\overrightarrow{\vec{x}(t_0)\vec{x}(t_1)}, \overrightarrow{\vec{x}(t_1)\vec{x}(t_2)}, \dots, \overrightarrow{\vec{x}(t_{n-1})\vec{x}(t_n)}.$$

The arclength of the curve S is then approximated by the sum of the length of these line segments

$$S \approx \sum_{j=1}^n \|\overrightarrow{\vec{x}(t_{j-1})\vec{x}(t_j)}\| = \sum_{j=1}^n \|\vec{x}(t_j) - \vec{x}(t_{j-1})\|.$$

Since $\vec{x}$ is differentiable, we also have the approximation

$$\vec{x}'(t_j) = \lim_{h \rightarrow 0} \frac{\vec{x}(t_j) - \vec{x}(t_j + h)}{h} \approx \frac{\vec{x}(t_j) - \vec{x}(t_{j-1})}{t_j - t_{j-1}},$$

and therefore

$$S \approx \sum_{j=1}^n \|\vec{x}'(t_j)\| (t_j - t_{j-1}).$$

Taking finer and finer approximations, the sum converges to an integral against dt and we recover the desired formula.

Example 23. Consider the helix parameterised by $\vec{x}(t) = (\cos t, \sin t, t)$, for $t \in [0, 2\pi]$.

Its tangent vector at each t is $\vec{x}'(t) = (-\sin t, \cos t, 1)$, and so the tangent line of $\vec{x}$ when $t = \pi$ has parameterisation

$$\vec{x}(\pi) + s \cdot \vec{x}'(\pi) = (-1, 0, \pi) + s \cdot (0, -1, 1), \quad \forall s \in \mathbb{R}.$$

Moreover, the speed of the curve is constant

$$\|\vec{x}'(t)\| = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{2}.$$

Thus, the arclength of the helix is

$$S = \int_0^{2\pi} \sqrt{2} dt = 2\sqrt{2}\pi.$$

Example 24. Consider the following two curves $\vec{x} : [0, 4] \rightarrow \mathbb{R}^2$, $\vec{x}(t) = (t, t)$ and $\vec{y} : [0, 2] \rightarrow \mathbb{R}^2$, $\vec{y}(t) = (t^2, t^2)$. We note that these two curves are two different parameterisations of the line segment joining the origin to the point $(2, 2)$ with their speeds always positive away from the end points. They should therefore give the same arc length value. Indeed

$$\begin{aligned} \int_0^4 \|\vec{x}'(t)\| dt &= \int_0^4 \|(1, 1)\| dt = \int_0^4 \sqrt{2} dt = 4\sqrt{2}, \\ \int_0^2 \|\vec{y}'(t)\| dt &= \int_0^2 \|(2t, 2t)\| dt = \int_0^2 2\sqrt{2}t dt = 4\sqrt{2}. \end{aligned}$$

Definition 1.7. A C^1 -curve $\vec{x} : (a, b) \rightarrow \mathbb{R}^n$ is called *regular* if

$$\|\vec{x}'(t)\| > 0, \quad \forall t \in (a, b).$$

Exercise: Prove that the following independence of parameterisation for the arc length:

If $\vec{x} : (a, b) \rightarrow \mathbb{R}^n$, $\vec{y} : (c, d) \rightarrow \mathbb{R}^n$ are two regular C^1 -curves with the same image

$$\{\vec{x}(t) \in \mathbb{R}^n : t \in (a, b)\} = \{\vec{y}(s) \in \mathbb{R}^n : s \in (c, d)\},$$

show that the arc length of $\vec{x}$ is equal to the arc length of $\vec{y}$.